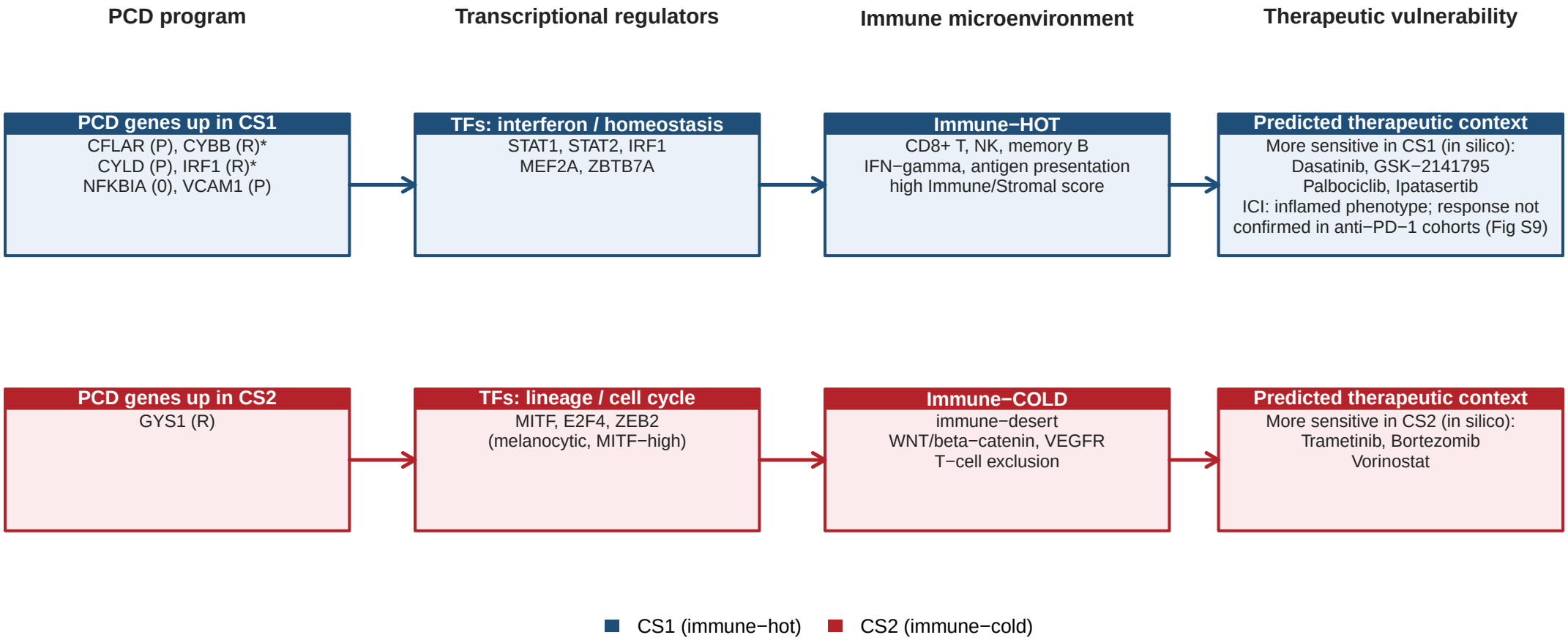
PCD dysregulation −> transcription factors −> immune state −> therapeutic response



Arrows denote directional associations inferred from regulon activity (Fig 3), immune deconvolution (Fig 4) and drug−sensitivity analyses (Fig S10); associative, not causal.

PCD genes placed by subtype expression (limma FDR < 0.05). (R)/(P)/(0) = risk / protective / zero weight in the CMPIS; * = multivariable weight opposes the gene's univariate HR; not subtype−specific: PIM2 (0).